J1A
DIN96 / male right-angle

ROW A

A1	
A2	
A3	GND
A4	+12V_BP
A5	+12V_BP
A6	GND
A7	+5V_BP
A8	+5V_BP
A9	GND
A10	
A11	
A12	
A13	
A14	
A15	
A16	HOST_SCL
A17	
A18	
A19	
A20	HOST_RX
A21	
A22	
A23	
A24	GND
A25	+5V_BP
A26	+5V_BP
A27	GND
A28	+12V_BP
A29	+12V_BP
A30	GND
A31	
A32	

J1B
DIN96 / male right-angle

ROW B

B1	GND
B2	GND
B3	GND
B4	+12V_BP
B5	+12V_BP
B6	GND
B7	+5V_BP
B8	+5V_BP
B9	GND
B10	GND
B11	GND
B12	GND
B13	GND
B14	GND
B15	GND
B16	GND
B17	GND
B18	GND
B19	GND
B20	GND
B21	GND
B22	GND
B23	GND
B24	GND
B25	+5V_BP
B26	+5V_BP
B27	GND
B28	+12V_BP
B29	+12V_BP
B30	GND
B31	GND
B32	GND

J1C
DIN96 / male right-angle

ROW C

C1	
C2	
C3	GND
C4	+12V_BP
C5	+12V_BP
C6	GND
C7	+5V_BP
C8	+5V_BP
C9	GND
C10	
C11	
C12	
C13	
C14	
C15	
C16	HOST_SDA
C17	
C18	
C19	
C20	HOST_TX
C21	
C22	
C23	
C24	GND
C25	+5V_BP
C26	+5V_BP
C27	GND
C28	+12V_BP
C29	+12V_BP
C30	GND
C31	
C32	

USBInterface

USB_VBUS	USB_VBUS
USB_D+	USB_D+
USB_D−	USB_D−
USB_CC1	USB_CC1
USB_CC2	USB_CC2
USB_VBUS_SENSE	USB_VBUS_SENSE

File: USBInterface.kicad_sch

Power

+5V_BP	+5V_BP
+12V_BP	+12V_BP
+3V3	+3V3
+5V_LOAD	+5V_LOAD
+12V_MOTOR	+12V_MOTOR
POWER_12V_SENSE	POWER_12V_SENSE

File: Power.kicad_sch

MotionController

+3V3	+3V3
RP_RUN	RP_RUN
RP_SWCLK	RP_SWCLK
RP_SWDIO	RP_SWDIO
USB_D+	USB_D+
USB_D−	USB_D−

File: MotionController.kicad_sch

Existing backplane interface: 12 V, 5 V, UART and I2C0.

No new pin assignments. All 24 V contacts remain unused.

J3
MOTOR 1

1	M1_BO1
2	M1_BO2
3	M1_AO2
4	M1_AO1

J4
MOTOR 2

1	M2_BO1
2	M2_BO2
3	M2_AO2
4	M2_AO1

J5
MOTOR 3

1	M3_BO1
2	M3_BO2
3	M3_AO2
4	M3_AO1

J13
ENDSTOP

1	ROT_ENDSTOP
2	+3V3
3	GND

MotionGPIO

+3V3	+3V3
ENC1_CS	ENC1_CS
ENC2_CS	ENC2_CS
ENC3_CS	ENC3_CS
ENC_MISO	ENC_MISO
ENC_MOSI	ENC_MOSI
ENC_SCK	ENC_SCK
HEATER_PWM	HEATER_PWM
HOST_RX	HOST_RX
HOST_SCL	HOST_SCL
HOST_SDA	HOST_SDA
HOST_TX	HOST_TX
M1_A1	M1_A1
M1_A2	M1_A2
M1_B1	M1_B1
M1_B2	M1_B2
M2_A1	M2_A1
M2_A2	M2_A2
M2_B1	M2_B1
M2_B2	M2_B2
M3_A1	M3_A1
M3_A2	M3_A2
M3_B1	M3_B1
M3_B2	M3_B2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM
POWER_12V_SENSE	POWER_12V_SENSE
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_CS	ROT_CS
ROT_ENABLE	ROT_ENABLE
ROT_ENDSTOP	ROT_ENDSTOP
ROT_MISO	ROT_MISO
ROT_MOSI	ROT_MOSI
ROT_SCK	ROT_SCK
TEMP_IN	TEMP_IN
TOOL_PWM1	TOOL_PWM1
TOOL_PWM2	TOOL_PWM2
USB_VBUS_SENSE	USB_VBUS_SENSE
VAC_PWM	VAC_PWM

File: MotionGPIO.kicad_sch

MotionDrivers

+3V3	+3V3
+12V_MOTOR	+12V_MOTOR
M1_A1	M1_A1
M1_A2	M1_A2
M1_AO1	M1_AO1
M1_AO2	M1_AO2
M1_B1	M1_B1
M1_B2	M1_B2
M1_BO1	M1_BO1
M1_BO2	M1_BO2
M2_A1	M2_A1
M2_A2	M2_A2
M2_AO1	M2_AO1
M2_AO2	M2_AO2
M2_B1	M2_B1
M2_B2	M2_B2
M2_BO1	M2_BO1
M2_BO2	M2_BO2
M3_A1	M3_A1
M3_A2	M3_A2
M3_AO1	M3_AO1
M3_AO2	M3_AO2
M3_B1	M3_B1
M3_B2	M3_B2
M3_BO1	M3_BO1
M3_BO2	M3_BO2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM

File: MotionDrivers.kicad_sch

J6
MOTION ENCODERS

1	GND
2	+3V3
3	ENC_MOSI
4	ENC_MISO
5	ENC_SCK
6	ENC1_CS
7	ENC2_CS
8	ENC3_CS

J12
ROTATION ENCODER

1	+3V3
2	ROT_MISO
3	ROT_MOSI
4	ROT_SCK
5	ROT_CS
6	GND

J14
ROTATION MOTOR

1	ROT_OUT1
2	ROT_OUT2
3	ROT_OUT3
4	ROT_OUT4

J19
TOOL PWM

1	GND
2	TOOL_PWM1
3	TOOL_PWM2
4	+3V3

ToolOutputs

+3V3	+3V3
+5V_LOAD	+5V_LOAD
HEATER_PWM	HEATER_PWM
HEAT_LOW	HEAT_LOW
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_ENABLE	ROT_ENABLE
ROT_OUT1	ROT_OUT1
ROT_OUT2	ROT_OUT2
ROT_OUT3	ROT_OUT3
ROT_OUT4	ROT_OUT4
VAC_LOW	VAC_LOW
VAC_PWM	VAC_PWM

File: ToolOutputs.kicad_sch

J15
HEATER + TEMP

1	HEATER_5V
2	HEAT_LOW
3	+3V3
4	GND
5	TEMP_IN

J16
VACUUM / 5 V

1	+5V_LOAD
2	VAC_LOW

J17
THERMAL CUTOFF

1	+5V_LOAD
2	HEATER_5V

J18
USB-C / SERVICE

VBUS	A4, B9 — USB_VBUS
CC1	A5 — USB_CC1
CC2	B5 — USB_CC2
D−	A7, B7 — USB_D−
D+	A6, B6 — USB_D+
SBU1	A8
SBU2	B8
SHIELD	SH
GND	A1, B12 — GND

J2
SWD / RP2350

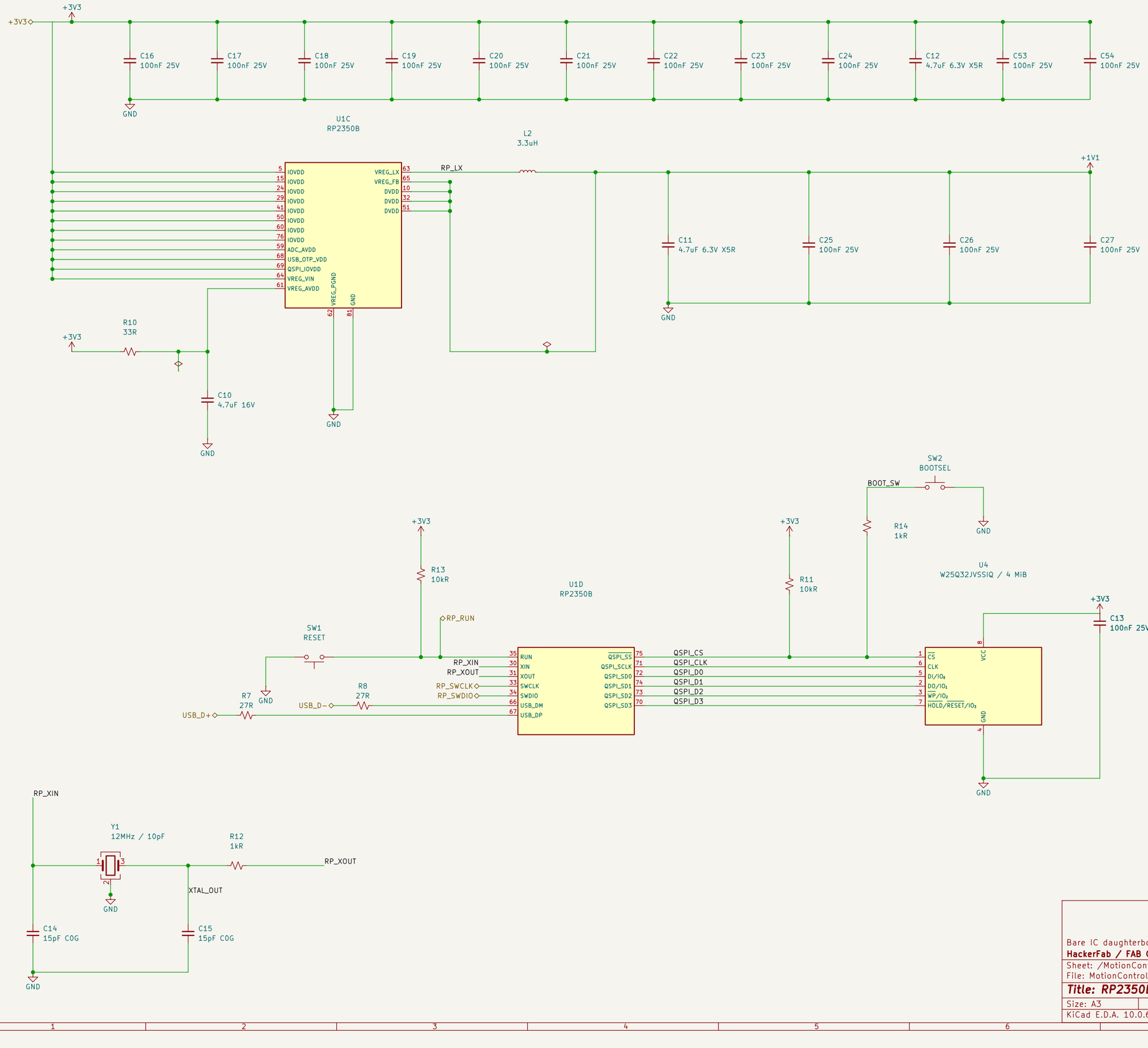
1	+3V3
2	RP_SWDIO
3	GND
4	RP_SWCLK
5	RP_RUN

USB: data only; power the backplane before service.

Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /
File: FabNanoDaughterboard.kicad_sch

Title: FAB Nano

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/7



Bare IC daughterboard / engineering review

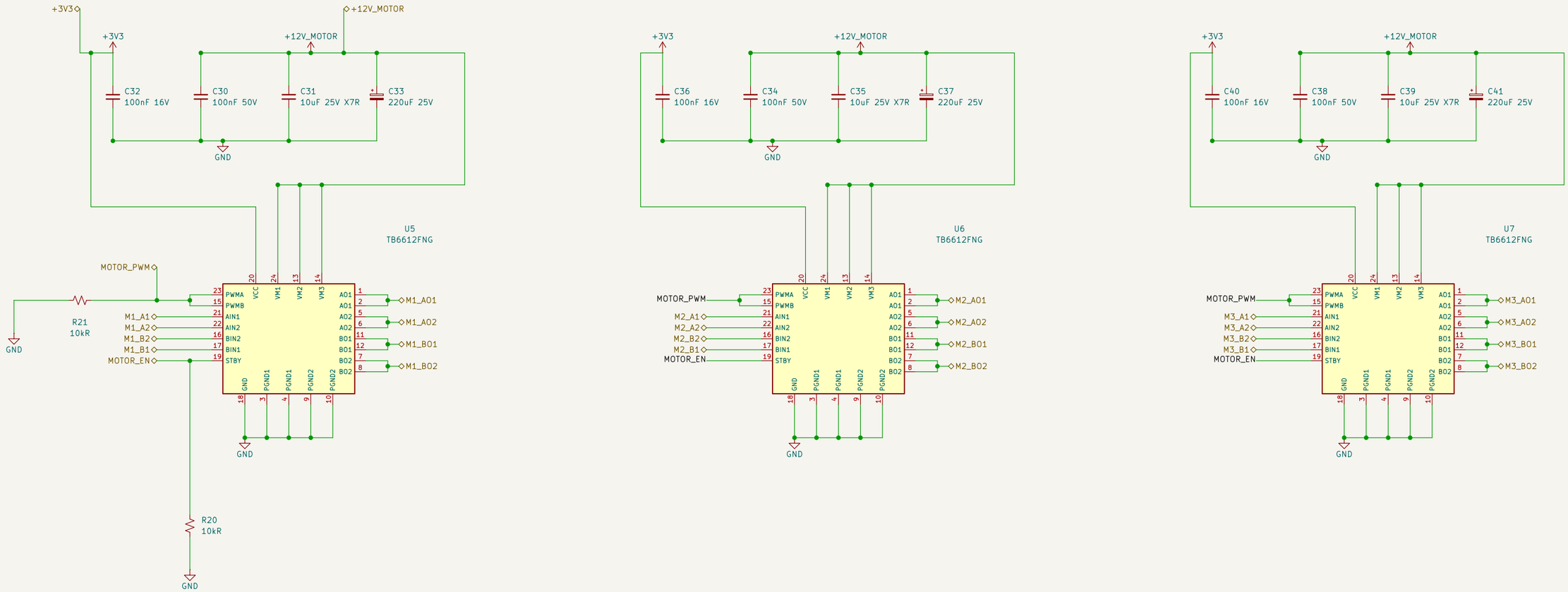
HackerFab / FAB Computer

Sheet: /MotionController/

File: MotionController.kicad_sch

Title: RP2350B

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/7



Bare IC daughterboard / engineering review

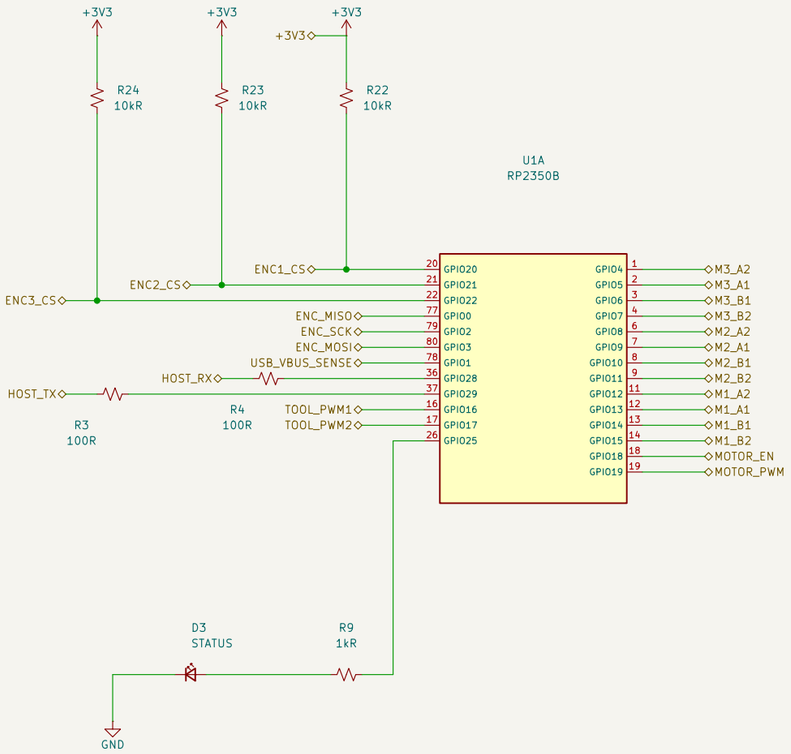
HackerFab / FAB Computer

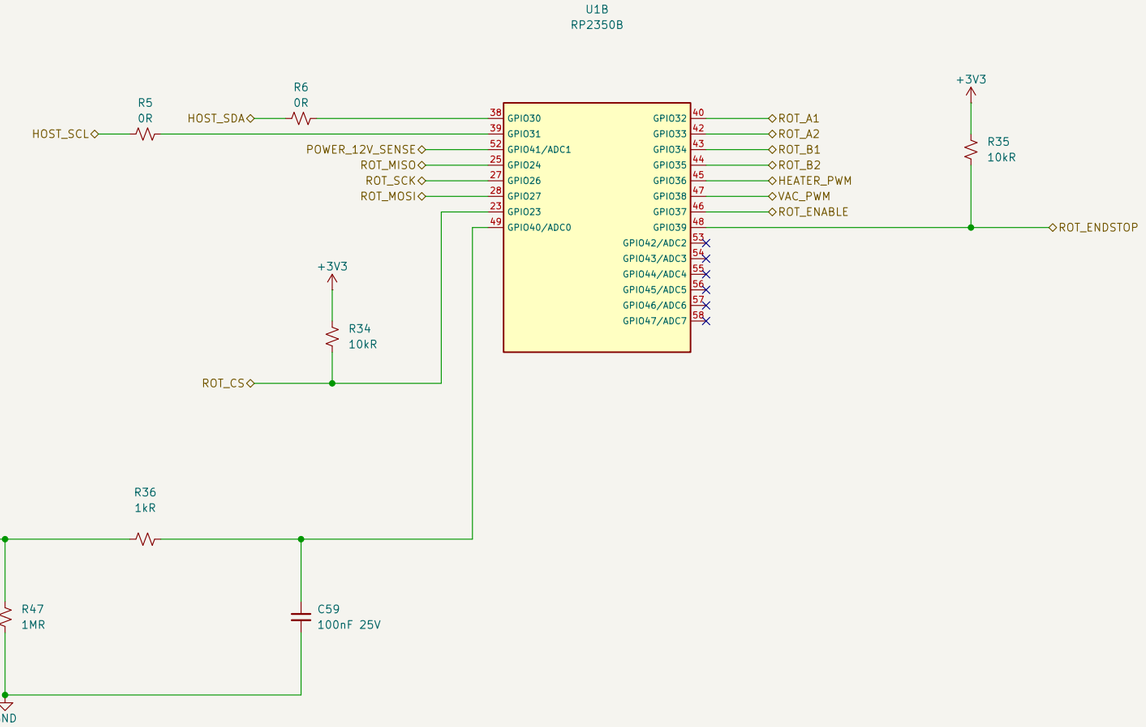
Sheet: /MotionDrivers/

File: MotionDrivers.kicad_sch

Title: Motion

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/7





One local controller: XYZ, rotation, temperature and vacuum.

Backplane UART and I2C0 contact assignments are unchanged.

Bare IC daughterboard / engineering review

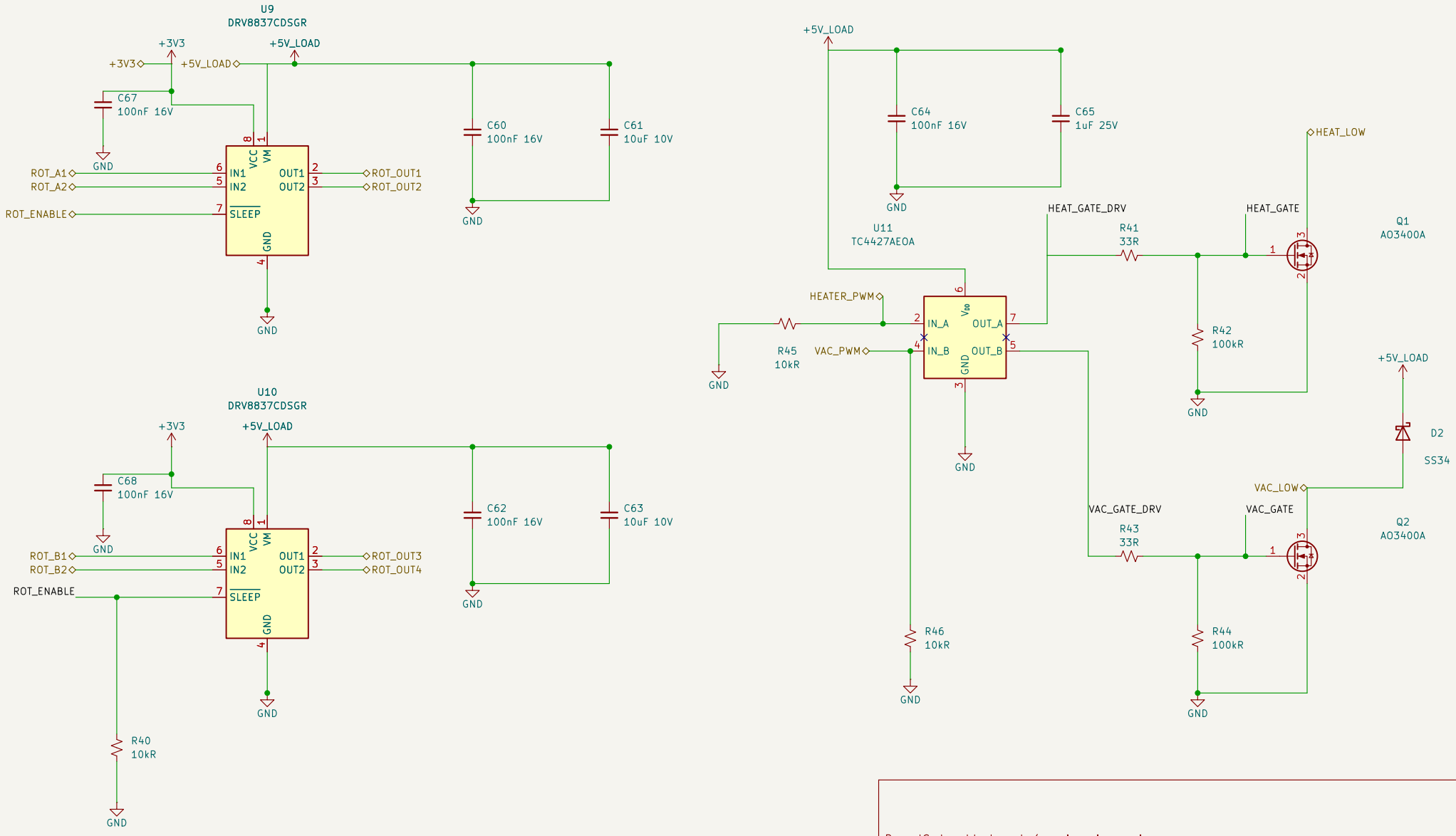
HackerFab / FAB Computer

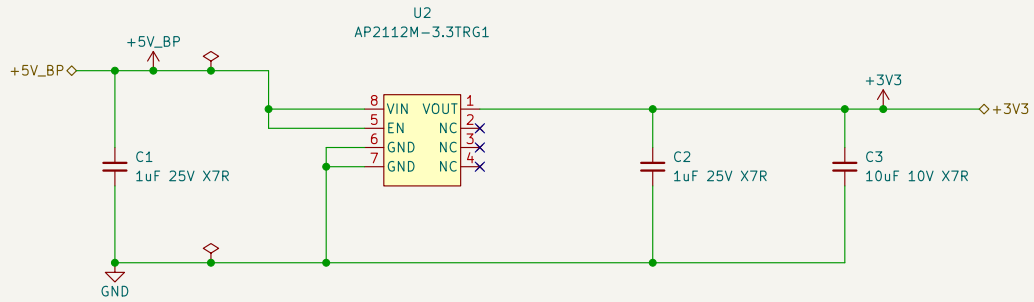
Sheet: /MotionGPIO/

File: MotionGPIO.kicad_sch

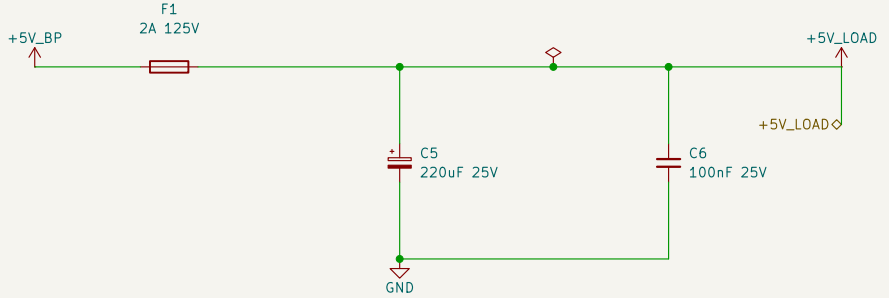
Title: RP2350B

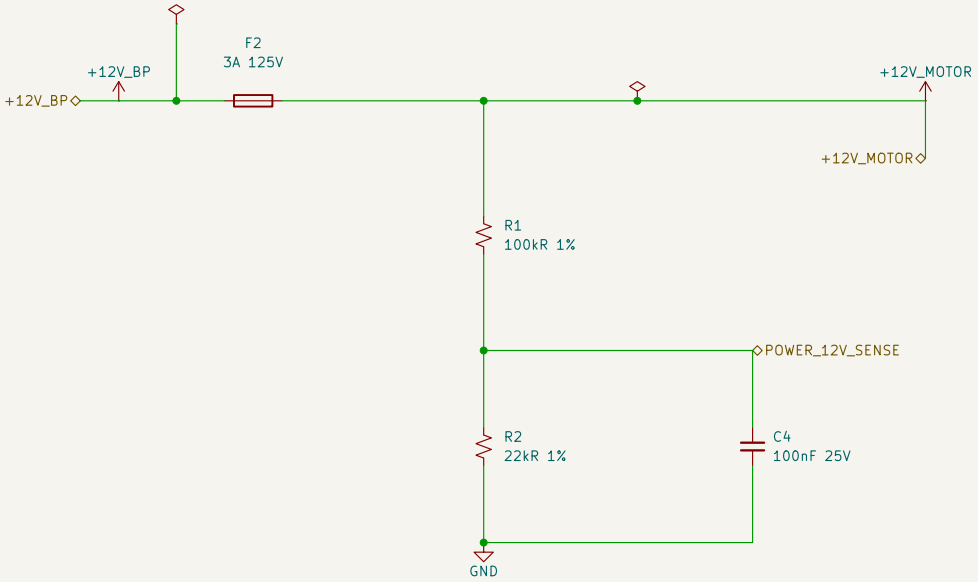
Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/7


U9
DRV8837CDSGR
+3V3
+5V_LOAD
C67
100nF 16V
C60
100nF 16V
C61
10uF 10V
GND
ROT_A1
ROT_A2
ROT_ENABLE
IN1
IN2
SLEEP
VCC
VM
OUT1
OUT2
ROT_OUT1
ROT_OUT2
U10
DRV8837CDSGR
C68
100nF 16V
C62
100nF 16V
C63
10uF 10V
ROT_B1
ROT_B2
ROT_OUT3
ROT_OUT4
R40
10kR
U11
TC4427AEOA
C64
100nF 16V
C65
1uF 25V
HEATER_PWM
VAC_PWM
IN_A
IN_B
OUT_A
OUT_B
R45
10kR
R46
10kR
HEAT_GATE_DRV
HEAT_GATE
R41
33R
R42
100kR
HEAT_LOW
Q1
AO3400A
D2
SS34
VAC_GATE_DRV
VAC_GATE
R43
33R
R44
100kR
VAC_LOW
Q2
AO3400A
Thermal cutoff J17 is in series with heater supply.
Headers and load wiring: top-level connector sheet.
Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /ToolOutputs/
File: ToolOutputs.kicad_sch
Title: Rotation, heater and vacuum
Size: A4
Date: 2026-11-06
Rev: A / DRAFT
KiCad E.D.A. 10.0.6
Id: 5/7



Logic only. No new backplane contact.





12 V absent: inhibit tool outputs.

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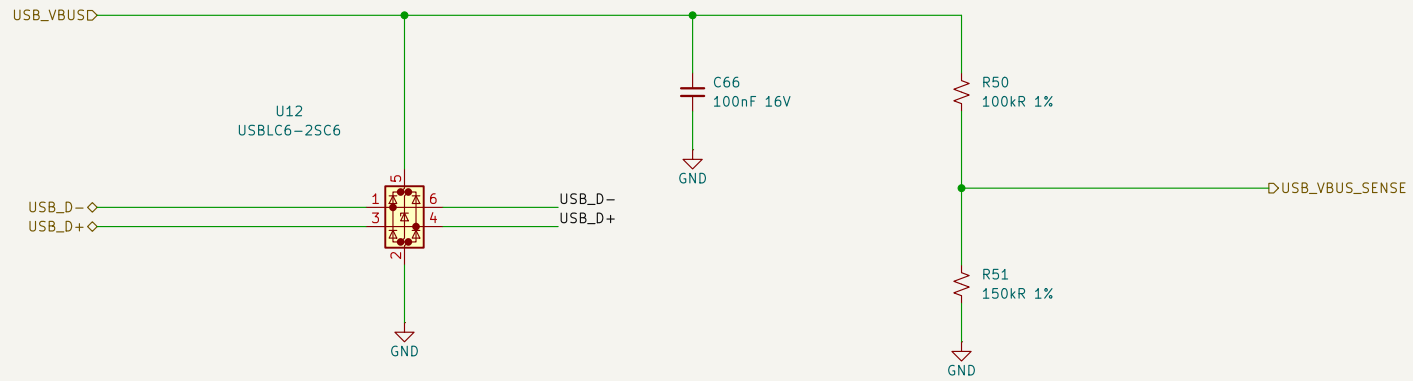
Sheet: /Power/

File: Power.kicad_sch

Title: Backplane rails and interfaces

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 6/7

USB SERVICE / ESD PROTECTION AND VBUS DETECTION



USB-C SINK / CC TERMINATIONS



Self-powered USB 2.0 service port. Backplane power must be present.

CC1 and CC2 have independent 5.1k pull-downs. VBUS feeds protection and sensing only.

Bare IC daughterboard / engineering review

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Sheet: /USBInterface/

File: USBInterface.kicad_sch

Title: USB service interface

Size: A4	Date: 2026-11-07	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 7/7